# WorldEdge: World Quality, Expansion and Task Execution

Evaluation design v5 — comparison protocol updated 20 October 2026 — all results pending

**Question.** Can the system deliver usable worlds, and can a fixed embodied executor complete tasks while those worlds grow without losing prior interaction state?

**Table 1: World utility and reliable incremental growth.** A compares complete generators on a common finite-world suite C. B compares growth strategies on suite G, with identical valid starting worlds, sequential requests and a fixed executor. Compare rows within a panel; do not average C and G. All rows are proposed evaluation configurations, not completed reproductions. *NE* = not evaluated; — = outside the stated protocol.

**A. Common finite worlds: generation quality and fixed-agent utility (C)**

| | World quality | | | | | | | | Agent task execution | |
|---|---|---|---|---|---|---|---|---|---|---|
| Generator | Load↑ | CNT↑ | ATR↑ | OORel↑ | OARel↑ | COL↓ | STB↑ | Reach†↑ | E2E $SR_C$↑ [95% CI] | $GC_C$↑ |
| SceneSmith [2] (candidate adapter) | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| EmbodiedGen V2 [6] (candidate adapter) | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| SceneCode [7] (candidate adapter) | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| EmbodiedSmith [9] (candidate adapter) | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| Code2Worlds-2 [10] (candidate adapter) | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| **WorldEdge** | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |

**B. Growing worlds: preservation, physical quality and cross-expansion tasks (G)**

| | World expansion and validity | | | | | Agent task execution | | Resources | |
|---|---|---|---|---|---|---|---|---|---|
| Strategy | $Expand_g$↑ | $Expand+Keep_g$↑ | Reach†↑ | $COL_\Delta$↓ | $STB_\Delta$↑ | **E2E $SR_G$**↑ [95% CI] | $GC_G$↑ | Time/seq (s) | Peak sim. RAM (GiB) |
| SceneSmith + regenerate/restore (candidate) | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| EmbodiedGen V2 + regenerate/restore (candidate) | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| Procedural append + normal persistence | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| Matched-component regeneration + restoration | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| **WorldEdge (full)** | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| w/o layout refinement | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| w/o preservation validation | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* | *NE* |
| *Offline reference; future requests are visible to the generator, not to the agent* | | | | | | | | | |
| Pregenerate + progressive reveal | — | — | *NE* | — | — | *NE* | *NE* | *NE* | *NE* |

**Metric lineage.** Semantic scores: SceneEval [1]; base physics definitions: SceneSmith [2]. Reach adapts PhyScene [3]; E2E SR/GC adapt task evaluation [5, 4]. Load accounting, Expand/Expand+Keep, affected-region populations and resource scope are specified here, not established standard metrics. Expansion/preservation evaluation already exists [15].
**Units and scale.** Scores are percentages; time and memory use the displayed units. Proposed main G setting: two starting rooms, four additions of one room each ($g = 4$); freeze after an independent feasibility pilot. C uses a common scale fixed by the adapter audit. Supporting curves use $g \in \{1, 2, 4, 8\}$. This measures finite tested scales, not unbounded reliability.
**Measurements.** A quality scores use all loaded worlds. In B, world diagnostics use fixed state probes on reset copies; agent scores use independent closed-loop executions. $Expand_g$ requires all $g$ additions to be valid; $Expand+Keep_g$ additionally requires every protected state to survive every boundary. Both use all scheduled diagnostic sequences, so Keep ≤ Expand. $COL_\Delta$/$STB_\Delta$ inspect additions and affected boundaries.
**Accounting and comparability.** E2E scores retain confirmed failures; unresolved cases use bounds. Disclose eligibility, assistance and sample sizes. B fixes the executor, seeds, critical assets, request information and budgets. External rows compare adapted systems; matched-component regeneration tests the update strategy. Expansion pauses the physics clock. Offline growth scores are inapplicable. Time includes retries; simulator RAM excludes remote memory.
**Supporting tables.** Full SceneEval diagnostics, global topology, conditional preservation, old-region drift, task-family breakdowns, reload-only controls, timeout rates and full generation costs accompany this table.
† Reach is a task-target geometric diagnostic, not executed navigation success. No overall world-quality composite or C+G aggregate is proposed.

# 1 What each panel establishes

**Panel A: complete finite-world systems.** The unit is a preregistered world request, with task specifications and $K$ executor seeds. Freeze the supported task intersection and adapter eligibility before measuring scores. Compare actual delivered worlds using the same simulator, physics settings, executor, perception, action implementation and budget. An adapter may perform declared format conversion; supplying missing articulation, layout repair or a persistent host changes the evaluated system and must be named. The five external rows are a shortlist, not a claim that all five already support the common interface. Unsupported configurations are disclosed and excluded before running the applicable comparison; they are not assigned zero.

C proposes navigation, local manipulation/articulation and cross-room strata under fixed quotas; common cross-room support is not yet established. Narrow the common set or report separately eligible subsets before observing results. All required objects, initial states and task constraints are frozen. A generator that fails an accepted request contributes a confirmed system failure; a generator may not simplify the task to improve its score.

The evidence levels differ: SceneSmith reports quantitative model-based robot execution; EmbodiedGen V2 cites policy-training and transfer benefits [2, 6, 8]. SceneCode provides articulated interaction demonstrations, without a reported task-success benchmark [7]. EmbodiedSmith optimizes plans and initial robot poses in its fixed-scene quantitative setting; Code2Worlds-2 uses contact-proxy control in its matched-prompt study [9, 10]. None of these results establishes that all candidate systems already share our executor. Training gains, human usability scores and reference-package validation rates cannot be copied into our E2E SR columns.

**Panel B: online growth strategies.** G starts each method from the same valid physical world, including the same established object identities and state. Each online method receives its current state through the same information interface, the same currently released expansion request, available assets and deadline. Future requests are hidden until their release. Unlike C, initial seed-world construction is not part of G's online-method score or cost. Publish the shared seed-world construction procedure and expense separately. End-to-end G includes growth failures and agent failures from the start of the scheduled growth-and-task sequence.

The proposed external rows use SceneSmith or EmbodiedGen V2 to generate the currently requested complete world, followed by a shared, validated state-restoration adapter. Name the adapter in each result; these are new configurations, not published benchmark scores. Audit supported exports and task intersections first. Unsupported adapters are not zero-scoring methods. Record native editing or generate-new-room-plus-append alternatives when supported; do not choose a deliberately weak adapter or tune one on the test set. Freeze adaptation effort and pilot-selected configurations before the main comparison. A general claim of outperforming existing systems requires externally sourced runnable comparisons, not only WorldEdge ablations.

Matched-component regeneration instead keeps WorldEdge's generation components and task-critical assets fixed and changes the update strategy. This is the primary mechanism contrast. External-generator rows answer a separate complete-system question; differences in generator quality must not be attributed solely to growth mechanics. The main protocol pauses physical time at growth boundaries and resumes after loading and validation; it does not test asynchronous hot updates during robot motion.

Accept seeds using frozen, method-independent initial physics and task-feasibility checks, never downstream performance of the compared methods. Publish candidate counts, rejections and provenance; WorldEdge-only seeds limit the inference to that seed distribution. To isolate growth mechanics, fix task-critical asset versions, scales, collision shapes, joints and physical parameters. If asset selection varies, interpret the result as a complete-strategy comparison that includes asset selection.

Procedural append uses a declared template/grammar and normal simulator persistence. Full regeneration receives the changed world state and a competent identity/state restoration procedure, using the same available generation components and assets where possible. Do not compare against a baseline deliberately denied serialization. The two WorldEdge ablations remove the named refinement or preservation-validation-and-repair component while retaining normal save/restore; their precise implementation mapping must be fixed before running. They are explanatory controls, not substitutes for the independent strategies.

Freeze the restoration implementation on independent pilot cases. Require a no-growth serialize–rebuild–restore check covering every scored state field and tolerance. Permit direct use of known semantic and asset IDs. Disclose unsupported fields and differences in generation components; remaining differences restrict attribution to the concrete configurations, not a pure growth mechanism.

An evaluator-owned identity ledger binds semantic IDs to asset provenance, geometry, collision shapes, scale, articulation, instance properties and physical state. Re-instantiation is valid when this declared equivalence is preserved one-to-one; reusing an ID on a substitute is not. Permit the restoration adapter to rebuild old assets and structure directly from the ledger. Freeze allowed changes to the new region and declared connection interfaces. Existing task-relevant obstacles, collision settings and physical parameters remain invariant outside that scope, as do robot and executor states under the pause/resume contract. This prevents a system from preserving poses while silently making the task easier.

Pregenerate + progressive reveal sees the full request sequence in advance and opens regions on the same schedule. It is an offline reference, not an assumed upper bound or a fair online-information competitor. Match final requested content, agent-visible information and task order; disclose any geometry differences. Include full pregeneration time in this reference's Time/seq and mark online growth metrics inapplicable. Do not give the agent privileged future observations.

**Main scale and task.** The proposed main scale is $g = 4$ additions, starting from two rooms and ending at six requested rooms. This is a design choice, not a measured capacity claim. Verify feasibility on separate pilot seeds, then freeze main scale, object-density bands, deadlines and sample sizes before evaluation. Report separate $g = 1, 2, 4, 8$ curves without pooling their scores. If the controller only supports a smaller common scope, revise the scope and claim before observing comparative results.

A qualifying G task requires a change before expansion and a later dependency on that changed state. Validate each task card counterfactually: retaining identity and action history but undoing the early state change must break a specified later dependency. For example: store a particular cup in a drawer and close it; create a connected room with a destination surface; return to that drawer, whose containment must persist until retrieval; retrieve the same cup and deliver it to the new room. This illustrates one expansion, not the full $g = 4$ sequence. Full G success requires all $g$ scheduled additions to be valid and all prescribed task dependencies to pass. Each added room must have an indispensable task role, and an early changed state must remain relevant after the final addition. The validator checks order, identity, physical outcomes, protected constraints and final dwell time. Spawned duplicates, pre-satisfied goals or skipping the initial action cannot satisfy the contract. Run reload-only controls separately from real spatial growth.

## 2 World measurements and their denominators

**Finite-world quality.** Load is successful simulator loads divided by all C world requests. CNT, ATR, OORel and OARel follow SceneEval's count, attribute, object–object and object–architecture definitions; OORel is not OptiScene's 2D overlap ratio [1]. Each score is satisfied annotated constraints divided by all constraints of that type within a scene, then macro-averaged across applicable loaded scenes. It is not a fraction of objects or candidate pairs. Evaluate after cleanup and retain constraints for missing requested objects. Deletion does not necessarily strictly lower every score. OOB and functional-side ACC remain diagnostic columns; ACC's best functional-side clearance is not global reachability.

Freeze the annotation vocabulary, matching/judge configuration and evaluator commit. For this proposed protocol, zero-annotation cases are prespecified N/A with counts; a failed or unmapped judgment is unknown, never an automatic pass. Compute from structured per-constraint outcomes, including confirmed missing-object failures. These explicit aggregation safeguards are an adaptation: upstream relationship code can mark unsupported relations as satisfied, while a fork's message-based parser can omit early-return failures [18, 19]. This audit does not establish which path produced published scores.

COL is the fraction of eligible objects participating in excessive interpenetration; STB is the fraction stable under a fixed gravity rollout. SceneSmith Appendix M.2.1 specifies penetration greater than 1 mm, a 5 s rollout at 1 ms steps, and stability below 1 cm translation and 0.1 rad rotation; it excludes rugs, welds wall/ceiling-mounted objects and uses V-HACD with up to 128 convex pieces [2]. Its public fork differs on the threshold boundary and exposes eligibility/aggregation differences [19].

Our frozen adaptation uses the strict paper thresholds, excludes rugs from both numerators and denominators, and excludes welded bodies from STB rather than granting trivial stability. Report welded and eligible counts. Fix support classification, decomposition and physics settings across methods; validate exported articulation separately. Average object fractions within scenes, then across scenes with eligible objects, retaining N/A and unknown counts. A changed simulator is a declared adaptation, not reproduction of the published score. A failed load has no observed physics score and never receives perfect collision freedom.

Reach is the fraction of required task destinations with a connected collision-free approach region from frozen robot spawns, under a declared footprint, clearance and joint configuration. Missing required targets count as unreachable. Average within each requested world/sequence, then across units. This adaptation is motivated by PhyScene's accessibility analysis [3]; it is not an action-execution measure or a guarantee of grasp success. Separate building/door connectivity and articulation swept-volume diagnostics explain failures.

**Growth and preservation: sequence-level rates.** Use $M_g$ preregistered diagnostic sequences, each with $g$ requested additions and a frozen legal state-probe schedule. These run on reset copies independently of the evaluated policy. Initial valid mutations may be installed by a disclosed state editor for a pure world-state probe; that does not count as physical agent execution. Check probe legality, pose/joint tolerances, identity and containment before applying growth. The probe vocabulary and held-out distribution are fixed before results.

Let $a_{jt}$ indicate that addition $t$ in sequence $j$ is delivered, simulator-loadable, structurally compliant and connected within budget. Let $p_{jt}$ indicate preservation of every protected pre-existing state predicate after that addition. Define

$$\text{Expand}_g = \frac{1}{M_g}\sum_{j=1}^{M_g}\prod_{t=1}^{g} a_{jt}, \qquad \text{Expand+Keep}_g = \frac{1}{M_g}\sum_{j=1}^{M_g}\prod_{t=1}^{g}(a_{jt}p_{jt}).$$

An early confirmed failure fails the sequence; unattempted suffixes are recorded, not presented as observed transitions. A probe-setup failure is logged separately and cannot become an unnoticed exclusion or preservation pass. If setup is unresolved rather than a confirmed system failure, retain an unknown outcome. Expand deliberately does not test preservation; therefore Expand+Keep cannot exceed Expand. Report conditional Keep given a valid complete expansion and its count in the supplement. This separates transition reliability from corruption once expansion succeeds.

At each boundary freeze the physical pause/continue rule. Ordinary dynamics are not state corruption. Check semantic IDs, poses, joint values, relevant velocities, containment and task predicates with declared tolerances; simulator-local indices and contact caches need not be identical. Preservation means retaining correctly changed state, not returning to the original scene. This specific protocol is proposed here; persistence and revisit measurements already have prior art [11, 12].

**Post-growth validity.** For G, compute Reach against all final requested destinations on the last valid snapshot, so an absent region fails its destination tests even if expansion stopped early. $\text{COL}_\Delta$ and $\text{STB}_\Delta$ cover new entities and pre-existing boundary entities selected by a fixed geometric rule; both members of a new–old collision are eligible. Aggregate checkpoint diagnostics within sequences, then across sequences with observations. Publish checkpoint and sequence coverage, including failed-load exclusions. These conditional diagnostics must be read beside Expand; no score is imputed for an unobserved checkpoint. Also publish whole-world and protected-old-region scores to prevent good averages from concealing localized corruption.

# 3 Task outcomes, resources and inference

**Fixed executor and independent success.** Freeze policy checkpoint/prompt/memory, embodiment, perception, action frequency, action implementation and execution budgets. Disclose privileged observations, scripted skills, assisted grasp, snapping and teleportation. A high-level skill interface is acceptable if the underlying execution is specified and shared; symbolic state transitions cannot be relabeled physical control. Evaluate trajectories independently of the generator, using the task/goal-progress distinction and dependency constraints established by ALFRED and PARTNR [5, 4]. World diagnostic probes and closed-loop agent trials are separate cohorts; their marginal rates must not be multiplied.

For suite $X \in \{C, G\}$, let $N_X$ be all scheduled episode slots, $S_X$ the verified full successes and $q_i \in [0, 1]$ the constraint-aware goal progress in slot $i$:

$$\text{E2E SR}_X = S_X/N_X, \qquad \text{GC}_X = N_X^{-1}\sum_i q_i.$$

For each slot freeze its complete goal set $H_i$ and weights before generation:

$$q_i = \frac{\sum_{h\in H_i} w_h z_{ih}}{\sum_{h\in H_i} w_h}.$$

Use equal weights by default; disclose any prespecified alternative. A goal indicator $z_{ih}$ passes only with its bound identity, order and terminal-state constraints. Unexecuted, ungenerated or unactivated goals after a confirmed failure remain in the denominator. Unobserved goals retain unknown bounds. List which goals each constraint invalidates and which global invariants gate E2E success only. This constraint-aware implementation is our adaptation, not an unchanged ALFRED or PARTNR score. Confirmed pre-execution system failures receive zero contribution. Confirmed mid-execution failures retain independently verified partial progress under the frozen identity, temporal and terminal-state rules; an observed failure is not automatically zero progress. Use preregistered predicate weights and per-episode aggregation; complete success also requires all invariants and temporal dependencies. No reward is given for targets already satisfied initially. Slots sharing a generated world are not independent samples. The primary endpoint is G E2E SR at the frozen main $g$; C E2E SR establishes finite-world utility. Do not average the two.

**Failures, uncertainty and coverage.** For each method publish requested sequences/worlds, loaded, probe-initialized, attempted transitions, executed episodes, successes, confirmed failures and unresolved cases. Report conditional success among executed episodes and conditional preservation with their denominators. After a frozen infrastructure retry rule, $U$ unresolved binary outcomes give identification bounds $[S/N, (S+U)/N]$, distinct from a

sampling confidence interval. For goal progress, known-progress sum $Q$ and $U_q$ completely unknown progress values give bounds $[Q/N, (Q+U_q)/N]$; independently verified per-slot bounds may tighten this interval. Unknown outcomes do not count as observed failures or justify a completed ranking.

Pair methods by independent world/task-sequence seeds and randomize method run order within hardware/batch blocks. Use paired, world/sequence-clustered 95% intervals, preserving nested executor repetitions and shared-template dependence where applicable. Preregister the main G mechanism contrast against matched-component regeneration plus restoration, and specify external-system contrasts separately; do not choose the weakest baseline after results. Determine sample size with an independent pilot and a precision/power target, not by treating simulator frames as samples. Stratify task family, expansion depth, density, layout and held-out asset/composition status.

**Resource measurements.** Time/seq is mean wall time spent on generation/expansion processing per scheduled G diagnostic sequence, including initialization within scope, validation, retries and failed attempts. It excludes agent execution time; online rows exclude their common seed-world construction while the offline reference includes full pregeneration. Declare those scope differences prominently. Report successful-sequence completion latency, attempted-transition p50/p95, timeout rate and budget separately; a deadline-capped failed attempt is not a measured completion time. A cheap early failure is not an efficiency win.

Peak sim. RAM is mean per-sequence peak aggregate resident host memory of the simulator and its scene-state workers, in GiB, on identical hardware and cache settings. It excludes remote-model and generation-process memory; full host/GPU peaks, model/API costs and asset/cache costs belong in the resource ledger. Include failures and retries, publish missing resource samples and distinguish cold/warm runs.

**Supporting evidence and claim boundary.** The main table answers world utility and growth reliability. Attach the full semantic/physics diagnostics, architecture/door graph coverage, preservation errors by state type, goal-family breakdowns, rollout failure taxonomy, reload controls, and growth-cost curves. Blind visual/faithfulness evaluation is needed if visual quality is a claimed contribution. Policy-learning benefit requires a separate common held-out evaluation and equal experience budgets.

The causal claim is limited to the preregistered seed distribution, executor and tested scale. Independent world probes provide mechanism diagnostics; their marginal association with task scores does not establish mediation. Component ablations estimate the total effect of the implemented component removal under this protocol.

**Validate the measurement before ranking systems.** Use independent positive/negative controls: no-op or failed-load expansion must not pass Expand; corruption of old poses, joints or containment may leave Expand intact but must fail joint preservation; substituted assets under old IDs, removed obstacles, disabled collisions, pre-satisfied goals and skipped required rooms must fail the relevant invariant or task checks. Conversely, legal growth, equivalent re-instantiation, changed simulator indices and permitted numerical tolerance must not be rejected. Verify missing-outcome accounting, retry deduplication and deterministic replay. Read loaded simulator state and action evidence rather than trusting generator success flags. Calibrate preservation tolerances independently of STB thresholds, blind-audit a held-out sample of trajectories, and disclose evaluator errors and coverage. A persistent append baseline may legitimately achieve perfect preservation; task utility, new-region quality and cost must still distinguish systems.

Oneira overlaps extensible actionable state and persistent interactions; Genie/SIMA and AgentOdyssey overlap generated-world agent execution and online expansion [11, 13, 14]. They belong in the related-work capability matrix, not automatically in a numerically comparable mesh-physics leaderboard. Being non-comparable under this particular controller does not erase their prior art. Current local component and symbolic results cannot fill this table; no new physical experiments were run for this revision.

# 4 Source map and newly checked evaluation precedents

| Columns / decision | Status | Source and limit of inheritance |
|---|---|---|
| CNT, ATR, OORel, OARel | Adapted implementation | SceneEval definitions [1]; explicit missingness/aggregation safeguards. Not its unchanged leaderboard implementation. |
| COL, STB | Adapted protocol | SceneSmith thresholds [2]; freeze engine and explicit eligibility, including welded-body handling. |
| Reach; $\mathrm{COL}_\Delta$, $\mathrm{STB}_\Delta$ | Adapted diagnostics | PhyScene [3] motivates accessibility; task-target and affected-region populations are specified here. |
| E2E SR, GC | Adapted outcomes | ALFRED/PARTNR [5, 4]; our denominator includes generation/growth and our constraints are explicit above. |
| Load, Expand, Expand+Keep; $g$, cost scope | Proposed protocol | Defined here. Related expansion/preservation evaluation exists [15]; these exact predicates and populations require validation. |

HappyWorld-Bench already evaluates spatial expansion and preservation in Table 12 and Appendix B.6 [15]. Its growth/coverage and view-quality measures are not exact simulator-state preservation; its embodied track uses prompt-conditioned video predictions (Section 5), not the shared closed-loop controller proposed here. This distinction does not establish novelty. WorldArena separately evaluates perceptual and downstream functional utility [16], so the world-quality/task split is not a new contribution. WorldWeave constructs persistent growing geometry for video generation [17]; its scene-planning agent is not evidence of a task-executing robot. The source audit is targeted, not exhaustive, and establishes neither priority nor comparative performance.

# References


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